

Multimodal Feature Learning with VAEs and GANs for CLIP-Synthesized Images

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Abstract—Recent advancements in multimodal fusion techniques and learning primarily Contrastive Language-Image Pre-training (CLIP), have shown remarkable capabilities to generate semantically rich image representations by combining both textual as well as visual features. Still, extracting and enhancing various features from CLIP-generated images is a somewhat unexplored domain. Our study introduces a new framework that uses Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) to generate feature extraction and evaluation from CLIP-generated images. This work focuses on creating a hybrid model that evaluates the feature interpretability and also bridges the gap between generative and contrastive learning paradigms by offering a unified approach for multimodal representation learning.

Index Terms—LIP, Variational Autoencoder (VAE), Generative Adversarial Network (GAN), Feature Extraction, Multimodal Learning, Latent Representations, LIP, Variational Autoencoder (VAE), Generative Adversarial Network (GAN), Feature Extraction, Multimodal Learning, Latent Representations.

I. INTRODUCTION

The technique of integrating data from several sources or modalities for improved artificial intelligence (AI) model performance is known as multimodal data fusion. Multimodal AI has evolved rapidly, resulting in a development of strong models that are capable of understanding and analyzing data from a variety of data modalities, including text, images, and audio. Contrastive Language-Image Pre-training (CLIP), which uses contrastive learning to align textual and visual representations in a common embedding space, is one of the most prominent advancements in this area. As a major advancement in bridging the gap between language and vision, CLIP has shown impressive performance in zero-shot learning, image classification, and cross-modal retrieval. Although CLIP-generated image embeddings are semantically rich, they often fail to provide the structured and separate representations required for applications such as clustering, anomaly detection, and fine-grained image analysis. This limiting factor results from CLIP's emphasis on aligning high-level semantic features

as compared to guaranteeing intricate, well-organized latent representations. Consequently, it is still very difficult to extract useful, task-specific information from these embeddings. Variational Autoencoders (VAEs) provide an effective method to deal with this problem. Generative models called VAEs are made to use input data to learn probabilistic, organized latent spaces. They are useful for capturing underlying data distributions and improving feature extraction procedures because of their capacity to provide smooth and continuous latent representations. Performance in subsequent applications can be enhanced by extracting more structured, differentiated, and relevant characteristics from CLIP-generated images by encoding them using a VAE.

The integration of VAEs for feature extraction from CLIP-generated image embeddings is investigated in this study. In order to improve data interpretation and usefulness in multimodal AI tasks, the main goal is to find out if VAEs can improve the quality and structure of extracted features. This research advances our knowledge of multimodal data fusion and seeks to provide more efficient ways to use generative models in intricate AI systems. The primary aim of this research is to explore how VAEs can enhance the process of extracting features from image embeddings produced by CLIP. The study specifically seeks to evaluate how effectively VAEs extract relevant and disentangled features from CLIP outputs.

By solving these goals, this research improves multimodal data fusion methods and provides insight into how to improve the interpretability and significance of features obtained from generative AI models.

II. LITERATURE REVIEW

The feasibility of developing a Modality-Shared CLIP (MS-CLIP) architecture is examined in a study that focuses on the number of transformer model parameters that can be shared between modalities (language and visual) during contrastive

pre-training. In order to optimize efficiency, it seeks to identify which sub-modules within each layer should be shared and which should be modality-specific. The MS-CLIP design incorporates two auxiliary modules that are specialized to lightweight modality namely Effective Parallel Branching and Early Specialization. [15]

By presenting the idea of approximate sufficient statistics and proving that near-minimizers of the contrastive pre-training loss are sufficiently adaptable to fit a variety of tasks, another study creates a theoretical framework to explain why contrastive pre-training performs well in a range of downstream tasks. Numerical simulations conducted in the study validate the theoretical findings, demonstrating that contrastively pre-trained transformers exhibit strong generalization performance across various multi-modal tasks, thereby confirming the effectiveness of the proposed framework. [7]

Another study aims to develop a technique for using a Conditional Variational Autoencoder (CVAE) in conjunction with contrastive learning (CL) to extract style features from unlabeled datasets. Without depending on class labels, this method seeks to separate style features from other mixed features, a major obstacle in unsupervised learning. [12] [8] The authors present a novel framework named Retrieval Augmented Contrastive Language-Image Pre-training (RA-CLIP) to solve the data-hungry nature of CLIP. By including online retrieval of appropriate image-text pairs from a reference set, a small amount of data that was retained during training, this technique improves the training process. The authors conducted experiments on 10 image classification datasets as well as 2 object recognition datasets. [10]

One of the another paper presents the Variational Autoencoder algorithm, focuses primarily on the challenges associated with the evaluation and interpretation of extensive biomedical data. In addition, the paper addresses several datasets, including molecular structures, genetic sequences, medical pictures, and EHR data, along with the difficulties of data scarcity and complexity in biomedical research [9].

For addressing various characteristics of fake news, a novel study introduces the Vae-Clip model for multimodal fake news detection, which integrates semantic data from text as well as images.

The model's ability to overcome difficulties with rumor feature extraction and cross-modal semantic information was demonstrated when it was evaluated on four datasets and reached a maximum accuracy of 96.3% [16].

A study by Yasutomi et al. introduces an innovative approach that combines Contrastive Learning (CL) methods with a Conditional Variational Autoencoder (CVAE) to extract style features from unlabeled data. The experiments show how effective the method is in extracting styles from datasets such as MNIST and Google Fonts, and the results show that the performance of style extraction is constant across different CL methods. [13]

Through the smooth integration of sequential variational autoencoder and contrastive learning, the authors presented different kind of Conditional Variational Autoencoder (DCL-

CVAE) framework for diversified picture captioning. [11]. By combining contrastive learning with auto-encoders, the research introduces a unique method for out-of-distribution (OOD) detection that makes training on unlabeled data efficient. On the CIFAR-100 dataset, experimental results show an average improvement of 5.8% over unsupervised algorithms and 11% over supervised OOD detectors. [6]

In order to improve the expressivity of latent coding distributions, it suggests an energy-based prior. The technique, which works with hierarchical VAEs, trains the reweighting factor through noisy contrastive estimation. Experiments on a variety of datasets, including MNIST and CIFAR-10, demonstrate notable gains in generative performance. For improved data creation, the suggested method is straightforward and flexible enough to accommodate various VAE designs. [2]

With the goal to represent enriched latent structures in datasets, the contrastive variational autoencoder (cVAE) is introduced in this study. To improve the salient latent characteristics, cVAE blends deep generative models with contrastive learning.

The algorithm has a run-time comparable to that of conventional VAEs and is robust to noise and dataset purity. [1]

To provide an overview of the key methodologies and datasets used in recent multimodal learning studies, Table I presents a structured summary of the most relevant literature.

III. METHODOLOGY

- 1) **Data Collection** We selected a collection of semantically rich text prompts that cover a wide range of visual concepts in order to guarantee a solid and varied dataset [3]. The Flickr8k dataset was utilized for curation of the text prompts. The dataset contains image captions which are semantically rich and concise. All the images were generated using these captions as the basis. The steps in the selection process include:

Diversity in Semantics:

To produce a range of meaningful visuals, the prompts covered a number of categories, such as objects, scenes, and abstract notions [5]

Balanced Representation:

To prevent bias, care was taken to incorporate prompts with a variety of linguistic patterns (such as single objects, intricate settings, or stylistic or emotional cues).

Validation:

A selection of prompts was examined for generating potential and clarity, eliminating inputs that were unclear or unduly basic. Cross-modal consistency can be systematically analyzed with the generated aligned image-text pairings that are based on the completed prompt set.

The completed set of prompts is what is used to make matched image-text pairs, which make it possible to analyze cross-modal consistency in a planned way.

- 2) **Generation of images with CLIP**

Using Stable Diffusion, guided by CLIP embeddings that were derived from the text prompts, a total of 2,000 images were generated. After splitting this dataset

TABLE I

SUMMARY OF VARIOUS RECENT STUDIES MENTIONING DIFFERENT ISSUES IN GENERATIVE MODELING AND MULTIMODAL FEATURE EXTRACTION. THE TABLE HIGHLIGHTS DEVELOPMENTS IN CLIP-BASED AND VAE-BASED ARCHITECTURES BY OUTLINING THE PROBLEM ADDRESSED, APPROACH, DATASETS USED, THE RELATED RESULTS WITH CORRESPONDING REFERENCES.

PROBLEM	METHODOLOGY	DATASET	RESULT	REF.
Challenge of extracting fine-grained style features from unlabeled data	The proposed method utilizes a Conditional Variational Autoencoder (CVAE) that is conditioned by a Contrastive Learning (CL) model.	The study utilized the MNIST dataset, and an original dataset based on Google Fonts	In the Google Fonts dataset, the bounding boxes (size and placement) of characters were mainly transmitted as styles, suggesting that the approach did not always learn font faces as styles. In contrast, styles such as slant, size, and thickness were successfully transferred for MNIST.	[12]
Detecting out-of-distribution samples in feature space	Combines contrastive learning with an autoencoder to enhance feature separation	CIFAR-10, SVHN	Outperforms traditional OOD detection methods in various benchmarks	[6]
Detecting fake news in multi-modal data	Integrates CLIP for semantic feature extraction and a VAE for rumor-style feature extraction	Multi-modal fake news datasets	Improved accuracy in detecting fake news compared to baseline models	[16]
Extracting style features from unlabeled images	Uses Contrastive Conditioned VAEs with mutual information constraints to isolate style features	Unlabeled image datasets	Achieves better disentanglement of style features compared to standard VAEs	[12]
Generating diverse image captions	Uses Conditional VAEs with contrastive learning to enhance caption diversity	MS-COCO and Flickr30K	Produces more diverse and contextually accurate captions	[11]
Examining the impact of including lightweight modality-specific auxiliary modules to accommodate specializations in each modality..	Early Specialization and Efficient Parallel Branching	YFCC100M and LAION-400M for pre-training both the MS-CLIP model and the vanilla CLIP model.	When pre-trained on the YFCC-100M dataset, the (MS-CLIP) technique reduces the amount of parameters required in the model and outperforms the vanilla CLIP by up to 13% relative in zero-shot ImageNet classification.	[15]
Improving generative performance of VAEs by addressing the prior hole problem	Uses contrastive learning to align the prior distribution with the approximate posterior	Image datasets	Improves sample quality and representation learning in VAEs	[2]
Enhancing salient feature extraction in latent space	Introduces a contrastive variational autoencoder (cVAE) to separate shared and dataset-specific variations	MNIST, CelebA	Achieves better separation and enhancement of key latent features	[1]

of images into train and test data, 1600 images were used for training and the remaining 400 for testing. The corresponding text-image pairs were stored for alignment and downstream use. Images are evaluated using CLIP-guided generative models to ensure high semantic alignment with text prompts. The process involves the following steps:

- Model Selection:** In this proposed study, we are leveraging CLIP model along with generative architecture such as CLIP + stable diffusion to produce high quality images conditioned on text.
- Generation Process: CLIP Embeddings:** To execute image synthesis process, text instructions were encoded into CLIP's joint embedding space. **Iterative Refinement:** Stable Diffusion optimized pixels for maximizing CLIP based similarity between the prompt and the generated images. **Post-Processing:** This step is performed to perform various techniques in order to minimize artifacts such as noise, distortions, etc.
- Data Storage:** Prompts and their corresponding images are stored as paired sample for alignment analysis.

By this approach, we ensure that the dataset captures fine relationships between text and visual representations, which is very crucial for evaluating multimodal consistency. Sample of generated images with their corresponding prompt is shown in the figure 1.

- Pre-Processing: Image Normalization Using Resizing and Pixel Value Scaling** To ensure consistency, all the images that were generated were pre-processed and resized to a uniform resolution of 128×128 pixels. For stability of training dynamics, we have normalized pixel intensities to a range of either [0,1] or [-1,1] based on the requirement of generative model.
- Optional Augmentations (Conditional on downstream tasks)** During training phase, we have applied random augmentations in order to improve generalization and preventing overfitting. **Spatial Transformations:** Various transformations like minor cropping (upto 5% of image dimensions) and random horizontal flipping (+50%/-50%) are carried out. **Color Perturbations:** To simulate real world, Saturation, brightness and contrast are adjusted.
- Variational Autoencoder and its Architecture:** To ensure discriminative feature learning and model the latent



Fig. 1. Sample of generated images with their corresponding prompt

structure of generated images, we have used a hybrid architecture of VAE and GAN that was modified to work with CLIP embeddings. In order to extract a meaningful latent space from the generated images, a VAE-GAN was implemented [14]. Each image was encoded into a latent Gaussian Distribution $q(z|x)$ with the help of an encoder. Then, the VAE's latent space was modified to incorporate a GAN discriminator. The discriminator aims to make sure that the learnt latent features are discriminative and follow a distribution distinct from the prior distribution, for example, standard normal distribution. Reparameterization technique serves to sample the latent distribution (mean and variance) obtained from the encoder. The decoder serves a dual purpose as the VAE's reconstruction mechanism and the generator for the GAN. The network has been trained end-to-end using KL Divergence loss for the VAE and adversarial loss, resulting in reconstructions of the generated images. Then, the decoder has been used to rebuild the images using the latent vectors. The various components of the design are as follows:

a) **Encoder Network**

- **Function:** It maps an input image x to its corresponding latent distribution represented as $q(z|x)$
- **Architecture:** Two parallel dense layers make up a neural network based on convolution (CNN) containing feature maps that have been gradually downsampled.
 - **Mean $[\mu]$** : It is calculated for predicting the latent gaussian distribution's centre.
 - **Log Variance $[\sigma^2]$** : It is used for predicting logarithmic variance in order to have stable training.

b) **Latent Space** The Encoder maps the generated input images to a latent space which is probabilistic in nature.

c) **Integrating GAN's Discriminator** Integration of GAN's along with VAE was inspired by [4].

- **Purpose** This process of integration is done in order to perform discriminator's operation directly on latent vector instead of operating on pixel space.
- **Design** The GAN's Discriminator is integrated with VAE, taking latent features as input and produces output by classifying whether those features are discriminative or non-discriminative. In our work, decoder itself is behaving as a generator for GAN architecture.
- **Loss**
 - i) **Reconstruction Loss:** Measures image reconstruction error.
 - ii) **KL Divergence:** Regularizes the latent distribution to match a standard normal distribution.
 - iii) **Adversarial Loss:** Trains the VAE to produce latent vectors indistinguishable from real ones, uses BCE Loss.

d) **CLIP Adaptation**

- **Feature Alignment:** The VAE's decoder is trained to recreate images whose CLIP embeddings match those of the input prompt in order to guarantee cross-modal consistency.

The architectural diagram of our methodology is depicted in figure 2.

IV. EXPERIMENT AND RESULTS

A. Initial Results

The initial training and evaluation of the VAE-GAN yielded the following results shown in Table II. With reconstruction accounting for a majority of the loss, the VAE is learning effectively. The latent space is moderately regularized, as shown by the low KL Divergence. Low Accuracy of the discriminator indicates that it is not yet able to consistently

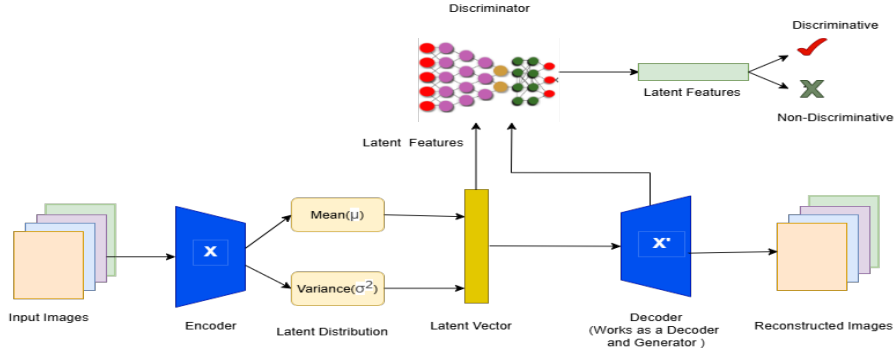


Fig. 2. Architectural diagram representing various components and their working.

TABLE II
VAE-GAN HYBRID MODEL: EVALUATION OF VARIOUS PARAMETERS.

Evaluation Parameter	Result
Reconstructiona Loss	6.22
KL Divergence Loss	0.83
Discriminator Loss	124.93
Discriminator Accuracy	50.3
Validation Loss	0.07
Validation Accuracy	0.528

distinguish between the generated and true latent vectors. With low loss and decent accuracy, the validation metrics show a slight improvement in generalization than training. The graph in the figure 3 and figure ?? represents the training losses, discriminative accuracy and ROC for our results achieved.

B. Results with Data Augmentation

When data augmentation is performed over the generated data, following results were obtained. The training losses, discriminative accuracy, and MSE for the same is represented by the figure 4.

C. Comparison with Baseline Models

The plot in figure 5 illustrates the comparison between the proposed VAE-GAN architecture and baseline VAE. The proposed architecture shows a rapid drop in loss, stabilizing significantly earlier than the baseline. In contrast, VAE continues to decrease gradually, indicating slower convergence and less efficient feature learning. This indicates that the representational capacity and better training dynamics of the VAE-GAN architecture, making it more effective for multimodal generation tasks

D. PSNR Evaluation

During training or validation, an example batch (Batch 89) achieved a Peak Signal-to-Noise Ratio (PSNR) of 12.3309dB. Given the low PSNR Value, the reconstructed images reveal a significant amount of degradation as compared to the original inputs. This suggests that the VAE Decoder is still unable to efficiently capture the detailed visual information from the latent space.

The model's sensitivity to batch size and training duration

was explored, with results indicating that smaller batches and longer training schedules (up to 100 epochs) led to marginal but consistent improvements in PSNR and SSIM values.

V. DISCUSSION

Our study demonstrates the importance of variational autoencoders while extracting meaningful latent representations from the images generated using the CLIP model. In addition, this study illustrates that the learned space representations generated by this method preserve significant semantic features. The performance and stability of the CLIP enhanced VAE-GAN model has shown almost no noticeable sensitivity to several hyperparameters, like the latent dimensionality, GAN Weight, learning rates, batch size and no. of epochs. We tried 3 variations of latent dimensions – 64, 128 and 256. Despite 256 being a very large dimension and 64 being very small, there was not a big difference on the performance. GAN weight still made a little noticeable influence, low values of weights led to blurry reconstructions, higher values led to noise, hence we concluded with gan-weight=0.1 as it gave the best trade-off between perceptual quality (SSIM, FID) and reconstruction fidelity (PSNR). Different batch sizes and learning rates didn't make much difference. However, increasing the number of epochs ensured that there was no underfitting, although 50+ showed a clear sign of plateau in training.

VI. CONCLUSION AND FUTURE WORK

In this paper, we presented a novel framework that leverages Variational AutoEncoders (VAEs) to extract latent feature representations from images generated via Contrastive Language-Image Pre-training (CLIP) models. By curating a diverse set of text prompts and using CLIP-guided generative models such as VQGAN and Stable Diffusion, we created a rich synthetic dataset that served as the basis for exploring representation learning in a controlled, semantically aligned setting. Future work will explore more advanced VAE architectures, such as hierarchical or β -VAEs, to further improve disentanglement and interpretability. Additionally, we plan to extend this framework to incorporate multimodal fusion with textual embeddings, enabling richer cross-modal understanding and opening new possibilities for tasks like visual question answering, sentiment analysis, and content moderation in digital media.

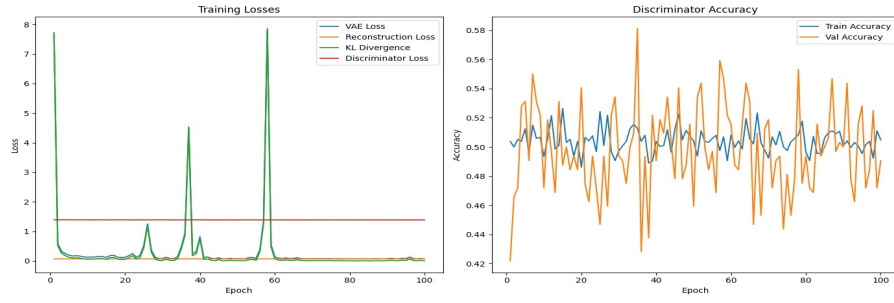


Fig. 3. Training Loss and Discriminative Accuracy before data augmentation



Fig. 4. Training Loss and Discriminative Accuracy and MSE after data augmentation

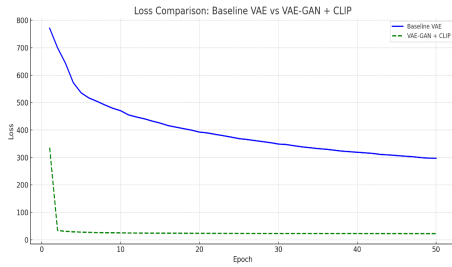


Fig. 5. Comparison between the proposed VAE-GAN architecture and baseline VAE

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